

ANGLO-CHINESE JUNIOR COLLEGE
DEPARTMENT OF CHEMISTRY
Promotional Examinations

CHEMISTRY
Higher 2

9729

Question Booklet

1 October 2018
2 hours

Additional Materials: Optical Answer Sheet
 Soft clean eraser
 Soft pencil
 Writing Paper
 Data Booklet
 Answer Booklet

READ THESE INSTRUCTIONS FIRST

Write your index number and name on all the work you hand in.
Write in dark blue or black pen **except** on the Optical Answer Sheet.
You may use a soft pencil for graphs and rough working.
Do not use staples, paper clips, highlighters, glue or correction fluids.

Answer **all** questions.

Section A

There are **fifteen** questions in Section A. For each question there are four possible answers **A, B, C** and **D**. Choose the **one** you consider correct and record your choice in **soft pencil** on the Optical Answer Sheet.

Read the instructions on the answer sheet very carefully.

Each correct answer will score one mark. No marks will be deducted for wrong answers.
Any rough working should be done in this booklet.

Section B and C

Questions for Section B are found in the Answer Booklet. Answer them in the Answer Booklet.
Answer questions in Section C on writing paper.

A Data Booklet is provided.
You are reminded of the need for good English and clear presentation in your answers.

At the end of the examination, fasten answers to Section C securely to the Answer Booklet.
The number of marks is given in brackets [] at the end of each question or part question.

This document consists of **9** printed pages, including this cover page.



Section A (15 marks)

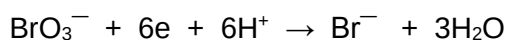
For each question, there are four possible answers, **A**, **B**, **C**, and **D**. Choose the **one** you consider to be correct and indicate your answers on the optical answer sheet provided.

- 1 Which particle will be deflected by the smallest angle in either direction in an electric field?

- A** ${}^7\text{Li}^+$
B ${}^{16}\text{O}^{2-}$
C ${}^{24}\text{Mg}^{2+}$
D ${}^{31}\text{P}^{3-}$

- 2 15.00 cm³ of 0.0200 mol dm⁻³ bromate(V), BrO_3^- , was found to react completely with 40.00 cm³ of 0.0150 mol dm⁻³ hydroxylamine, NH_2OH .

BrO_3^- ions are reduced as follows:



Which could be the half-equation for the oxidation of hydroxylamine?

- A** $\text{NH}_2\text{OH} + \text{H}_2\text{O} \rightarrow \text{NO}_2 + 5\text{H}^+ + 5\text{e}^-$
B $\text{NH}_2\text{OH} + 2\text{H}_2\text{O} \rightarrow \text{NO}_3^- + 7\text{H}^+ + 6\text{e}^-$
C $\text{NH}_2\text{OH} + \text{H}_2\text{O} \rightarrow \text{NO}_2^- + 5\text{H}^+ + 4\text{e}^-$
D $\text{NH}_2\text{OH} \rightarrow \text{NO} + 3\text{H}^+ + 3\text{e}^-$

- 3 Which statement about the kinetic theory of gases is correct?

- A** Collisions between the gas particles are inelastic.
B The volume of the gas is negligible.
C The average kinetic energy is the same for all gases at a given temperature, regardless of the identity of the gas.
D The kinetic energy of a gas particle is proportional to the temperature (in terms of degrees Celsius) of the gas.

- 4 Pyrogallol is a white crystalline powder that is used to absorb oxygen.

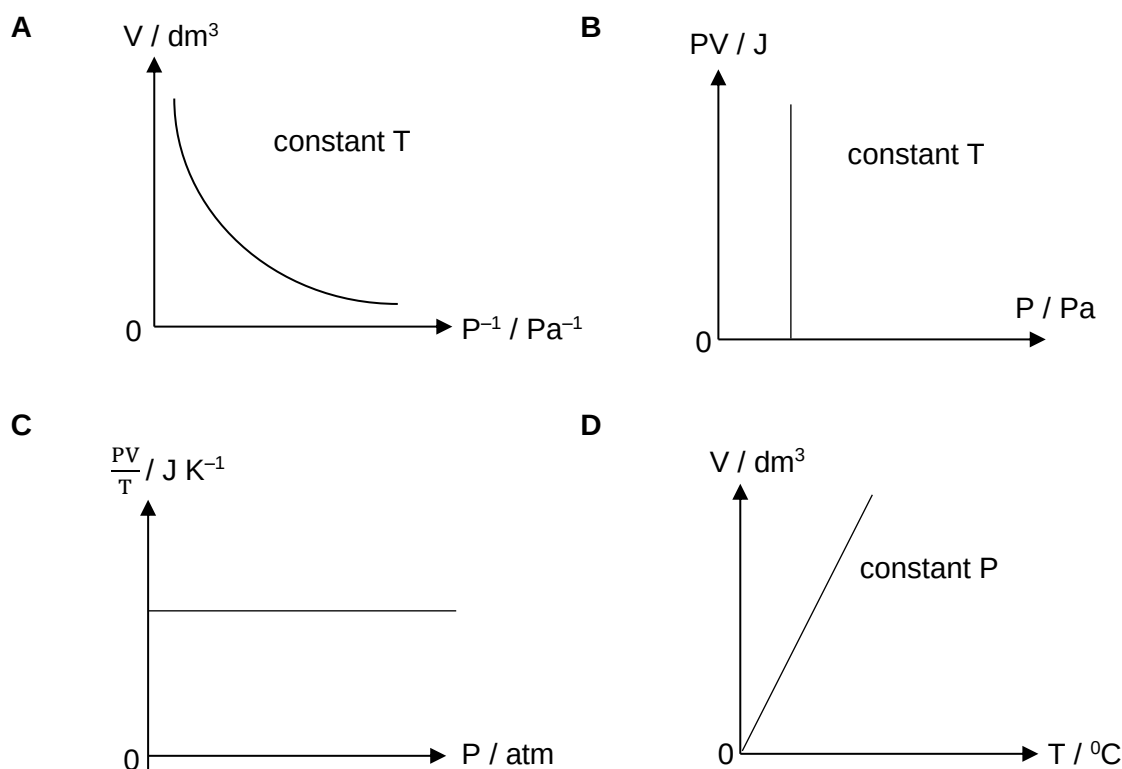
When 10 cm³ of a gaseous hydrocarbon is exploded with 200 cm³ of oxygen, there is a contraction to 190 cm³.

On further treatment with pyrogallol, there was a reduction of 160 cm³.

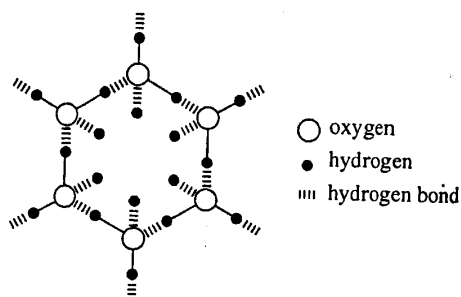
What is the molecular formula of the hydrocarbon? (All volumes are measured at room temperature and pressure).

- A** C_2H_4 **B** C_3H_4 **C** C_2H_6 **D** C_3H_6

5 Which diagram describes the behaviour of a fixed mass of an ideal gas?



6 The diagram shows part of the structure of ordinary ice.



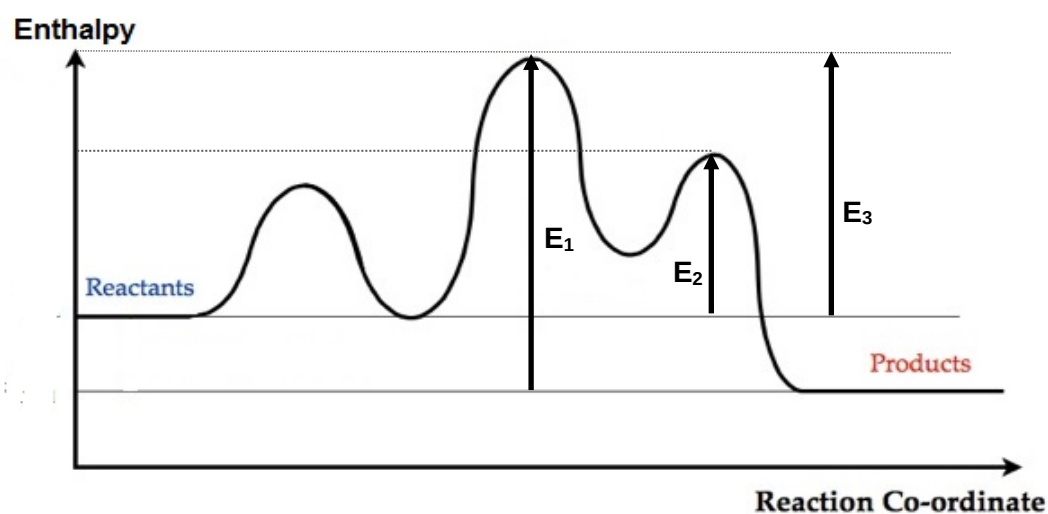
Which statement is correct?

- A** All the bond angles surrounding each oxygen atom are 120° .
- B** Four electrons from each oxygen are involved in forming hydrogen bonds.
- C** The hydrogen bonds, shown by the dotted lines, are stronger than the O-H covalent bonds.
- D** The open structure of ice causes ice to be denser than water.

- 7 $(\text{CH}_3)_2\text{S} \cdot \text{BCl}_3$ is a solid that is commonly used in laboratories as a convenient source of BCl_3 . When heated, it reversibly decomposes to $(\text{CH}_3)_2\text{S}$ and BCl_3 .

Which statement is true?

- A The C-S-C bond angle decreases when the solid decomposes.
 B The dative bond is from boron to sulfur.
 C The dative bond is formed using the 2p orbitals of boron and sulfur.
 D $(\text{CH}_3)_2\text{S}$ and BCl_3 act as the Lewis acid and Lewis base respectively in the formation of $(\text{CH}_3)_2\text{S} \cdot \text{BCl}_3$.
- 8 Which sequence is correct in terms of increasing radius?
- A $\text{Se}^{2-} < \text{Sr}^{2+} < \text{Rb}^+ < \text{As}^{3-}$
 B $\text{As}^{3-} < \text{Se}^{2-} < \text{Sr}^{2+} < \text{Rb}^+$
 C $\text{Rb}^+ < \text{Sr}^{2+} < \text{As}^{3-} < \text{Se}^{2-}$
 D $\text{Sr}^{2+} < \text{Rb}^+ < \text{Se}^{2-} < \text{As}^{3-}$
- 9 The energy profile diagram below represents a certain three-step reaction.



Which statements are correct about the above-mentioned reaction?

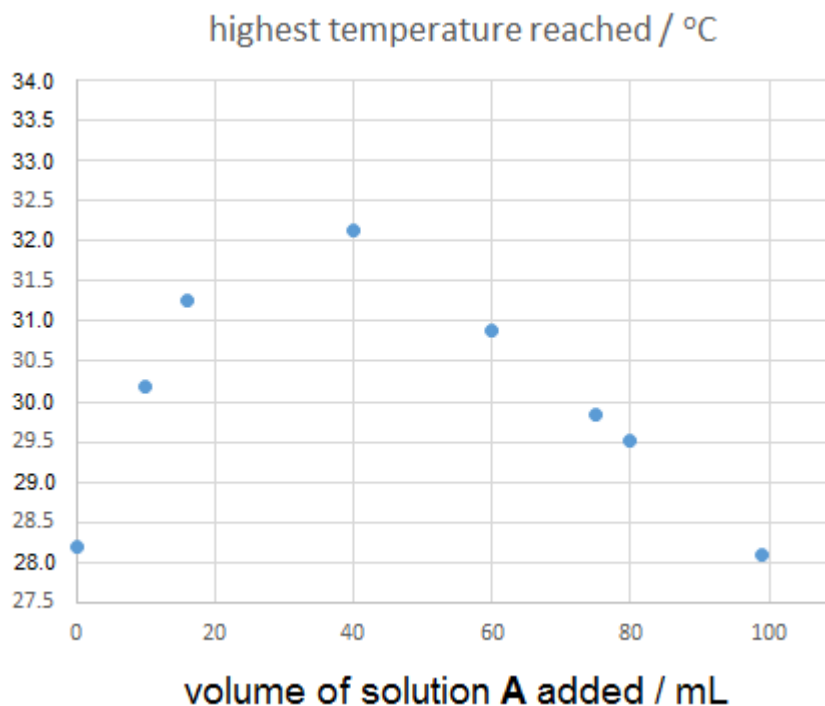
- 1 E_3 is the activation energy of the second step.
 2 ΔH can be obtained by subtracting E_2 from E_1 .
 3 There are equal number of intermediates and transition states.
- A 1 only
 B 2 only
 C 2 and 3 only
 D 1 and 3 only

- 10 **A** and **B** react exothermically.

A calorimetric experiment was conducted to investigate this reaction.

Solutions of **A** and **B** were at the same initial temperature.

A graph of the final temperature reached by mixing both solutions to a total volume of 100 mL is given below.



In what mole ratio does **A** and **B** react?

- A** 3:1
- B** 1:3
- C** 3:2
- D** 2:3

- 11 A certain first order reaction took 42 minutes for the initial concentration of the reactant to decrease from 0.48 mol dm^{-3} to 0.06 mol dm^{-3} .

How long will it take for the concentration of the same reactant to decrease from 0.40 mol dm^{-3} to 0.10 mol dm^{-3} ?

- A** 28 minutes
- B** 30 minutes
- C** 35 minutes
- D** 42 minutes

- 12 The kinetics of the reaction between **A** and **B** is studied through the initial rate method.

Initial concentrations / mol dm ⁻³		Initial rate (arbitrary unit)
[A]	[B]	
0.1	0.1	0.1
0.2	0.2	0.2
0.1	0.2	0.2

Which statements are correct about the reaction?

- 1 **A** is not part of the rate-determining step.
 - 2 The value of rate constant is 1.
 - 3 The unit of rate constant is mol dm⁻³ min⁻¹.
- A 1 only
 B 1 and 2
 C 1 and 3
 D 2 and 3

- 13 Which mixture will end up with a buffer solution for which the pH is below 7 at 25 °C?

- A 25.0 cm³ of 0.250 mol dm⁻³ NaOH and 25.0 cm³ of 0.150 mol dm⁻³ H₂SO₄
 B 25.0 cm³ of 0.100 mol dm⁻³ NH₃ and 25.0 cm³ of 0.200 mol dm⁻³ NaOH
 C 25.0 cm³ of 0.300 mol dm⁻³ HCl and 25.0 cm³ of 0.300 mol dm⁻³ NH₃
 D 25.0 cm³ of 0.200 mol dm⁻³ CH₃CO₂H and 25.0 cm³ of 0.100 mol dm⁻³ KOH

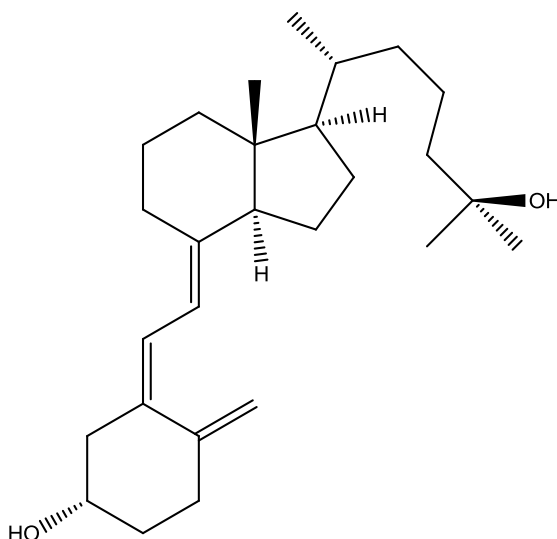
- 14 The table below shows the numerical values of the solubility products (measured at 25 °C) for some salts.

Salt	CdCO ₃	CoCO ₃	Ag ₂ S	CuS
K _{sp}	1.0 × 10 ⁻¹²	1.0 × 10 ⁻¹⁰	8.0 × 10 ⁻⁵¹	8.0 × 10 ⁻³⁷

Which statement is correct?

- A The solubility of these four salts will be increased at lower pH.
 B CuS is more soluble than Ag₂S.
 C CoCO₃ is less soluble than CdCO₃.
 D The K_{sp} value of CuS will decrease as less of it can dissolve when copper(II) nitrate is added to a saturated solution.

- 15** Calcifediol ($C_{27}H_{44}O_2$) is formed from vitamin D3 in the human liver. Doctors typically measure the amount of this compound to determine a patient's vitamin D levels.



Which statements are true about calcifediol?

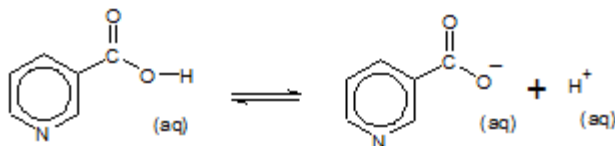
- 1** There are six chiral carbons.
 - 2** There are 21 sp^3 hybridised carbon atoms.
 - 3** All the $C=C$ bonds show cis-trans isomerism.
- A** 1 only
B 2 only
C 1, 2 and 3
D 1 and 3 only

END OF SECTION A

Section C (25 marks)

Answer **all** questions on foolscap paper.

- 1 Nicotinic acid is a form of vitamin B3, an essential human nutrient. Insufficient nicotinic acid in the diet can cause nausea, anemia, headaches, and tiredness.



Nicotinic Acid

It is given that the K_a value for nicotinic acid at 25°C is 1.4×10^{-5} .

- (i) A student pipettes out 25.0 cm³ of 0.1 mol dm⁻³ nicotinic acid (aq) and transfers it into a conical flask. Calculate the pH of this solution. [1]
- (ii) He then titrates the solution with 0.2 mol dm⁻³ NaOH (aq) using a suitable indicator. Calculate the pH of solution when endpoint is reached. [3]
- (iii) Calculate the pH at maximum buffer capacity. [1]
- (iv) Calculate the maximum pH of solution when NaOH(aq) is added in excess. [1]
- (v) Suggest a suitable indicator and state its colour change at endpoint. [1]
- (vi) Sketch the titration curve. Indicate all points calculated from (a)(i) to (iv) on the curve. [1]

[Total: 8]

- 2 The milk of magnesia is a short-term medication used to treat constipation, heartburn and indigestion. Its active ingredient is magnesium hydroxide, Mg(OH)₂.

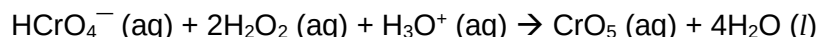
Mg(OH)₂ is a sparingly soluble salt with a solubility product of 5.61×10^{-12} mol³ dm⁻⁹ at 25 °C.

- (a) Calculate the solubility of Mg(OH)₂ in water at 25 °C. [1]
- (b) Calculate the solubility of Mg(OH)₂ in a 0.2 mol dm⁻³ magnesium nitrate solution at 25 °C. [2]
- (c) Calculate the minimum pH needed to precipitate out Mg(OH)₂ from a magnesium nitrate solution to the extent that the resulting [Mg²⁺] is no higher than 1×10^{-7} mol dm⁻³ at 25 °C. [2]

[Total: 5]

- 3** Chromium(VI) peroxide or chromium oxide peroxide (CrO_5) is a more powerful oxidising agent than potassium dichromate(VI).

It is a highly unstable compound formed by the addition of acidified hydrogen peroxide solutions to solutions of metal chromates or dichromates, such as sodium chromate or potassium dichromate.



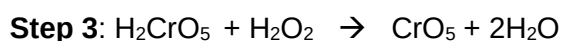
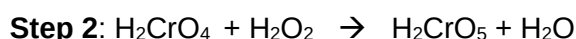
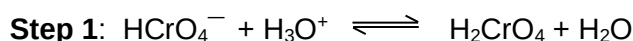
The generally yellow chromates or orange dichromates turn dark blue as chromium(VI) peroxide is formed.

- (a) Explain why the oxidation state of chromium cannot be +10, as what the molecular formula CrO_5 suggests. [1]
- (b) The kinetics of this reaction have been studied, and the rate equation is found to be

$$\text{Rate} = k [\text{HCrO}_4^-] [\text{H}_3\text{O}^+] [\text{H}_2\text{O}_2]$$

- (i) State the units of k , the rate constant. [1]

One of the proposed mechanisms for this reaction is as follows



- (ii) Suggest a certain physical property that can be monitored over time, in order to follow the progress of the reaction. [2]
- (iii) Show that step 2 is the rate determining step. [2]

[Total: 6]

- 4** Pentenes are components of gasoline and are often produced as by-products of thermal cracking of petroleum. Pent-2-ene is one of the isomers of pentene. It has the following formula: $\text{CH}_3\text{CH}_2\text{CH}=\text{CHCH}_3$.

- (a) Draw and label the two structures of pent-2-ene that show cis-trans isomerism. [2]

One constitutional isomer of pent-2-ene is the 1,2-dimethylcyclopropane.

- (b)(i) Draw and label the two structures of 1,2-dimethylcyclopropane that show cis-trans isomerism. [2]
- (ii) For each of the two structures drawn in (b)(i), explain whether it is optically active. [2]

[Total: 6]

END OF PAPER